J-space Probes on Gemma-2-9B-it

Method and results: where the collusion signal does and does not live

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NARCBench Transfer-stable lane

andylikescodes/narcbench-core-demo, branch jspace-separability

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Summary

A linear probe on the hidden states of google/gemma-2-9b-it tells colluding committee agents from honest ones. We asked whether the part of the hidden state the probe uses lies inside J-space, the directions that feed the model's next-word output, or outside it. To answer, we split every hidden state into a J part and a remainder, trained the same probe on each part, and tested on scenario families the probe had never seen. The first version of this report (4 November) used two stand-ins for J-space built from the output rows; this revision adds the construction of Gurnee et al. [3] itself, run on 5 November with the published Jacobian lens for this model, together with the dimension-matched controls the first version lacked.

1. A probe restricted to the output-row frames does not transfer. Trained on the Core scenarios and tested on 72 episodes from six new families, it scores 0.602 [0.471, 0.733] with the top-25 output rows and 0.549 [0.416, 0.683] with a sketched Jacobian (AUROC with 95% intervals; neither is distinguishable from chance), against 0.902 for the unrestricted probe. With the controls now run on Gemma, the output-row frame sits at the level of a random fixed 25-dimensional subspace (0.647; paired difference −0.045 [−0.091, +0.003]): what these frames hold on Core is no more than any 25 directions hold.
2. A probe restricted to the remainder transfers as well as the unrestricted probe, under every J-space estimate: 0.898 and 0.891 for the two frames, 0.891 [0.850, 0.927] for the paper's J-space, against 0.902.
3. A probe restricted to the paper's own J-space transfers in part. With the published Jacobian lens and the paper's sparse nonnegative pursuit (25 atoms per hidden state, about 1% of its variance), the J-space probe scores 0.755 [0.709, 0.796] on the 72 new-family episodes: above the random 25-dimensional subspace by +0.108 [+0.052, +0.160], above another turn's atoms from the same dictionary (0.559) by +0.196 [+0.136, +0.252], and below its complement by 0.136. Within the new families it reaches 0.854 [0.820, 0.887]. The result holds under eight registered variants of the decomposition (0.721 to 0.811); the only choice that moves it is the paper's step of filling the early-stopped pursuit to exactly 25 atoms, without which the J-space probe scores 0.811 [0.761, 0.856]. The transferable part sits at layers 22 and 23 and in the agents' final statements; on discussion turns of the new families the J-space probe is at the random level.

By the reading rules fixed in the pre-registered design, the transferable colluder signal is not in the output-row frames, and the paper's J-space carries a part of it while the complement carries most. Two things to keep in mind: the published lens was fitted on WikiText prompts

of 128 tokens and is applied here at the last token of long chat turns, outside its fitting range; and a probe on a one-percent-variance component reaching 0.85 to 0.94 at layers 22 and 23 says the signal there sits in a low-variance direction inside the J-lens cone, not that the model would verbalize it. The rest of this document is the method, step by step, with the exact settings behind every number.

1 What was measured

NARCBench [9] is a benchmark of committee games. Four language-model agents review a briefing and vote; in half of the episodes two of them secretly coordinate. The benchmark's detector is a linear probe: a direction in the model's hidden state that scores each agent, with episode-level statistics on the four scores. Gemma-2-9B-it generated all the episodes used here and is the model whose hidden states were probed.

J-space is the idea of Gurnee et al. [3]: at each layer, a small set of directions in the hidden state are the ones the rest of the network turns into output words, and the paper argues that these directions behave like a workspace the model can report from. If the collusion signal sat in those directions, a monitor could read it out as words. We tested this by restriction: project each hidden state onto its J-space directions (the J-only arm) or onto everything else (the complement arm), run the benchmark's probe on each, and compare with the probe on the full hidden state. The comparison that matters is on new scenario families, because inside the training distribution almost any restriction works.

2 Data

Model. google/gemma-2-9b-it [2]: 42 decoder blocks, hidden width $d = 3{,}584$, vocabulary $V = 256{,}000$. Input and output embeddings are tied, so the output matrix $W_U \in \mathbb{R}^{V \times d}$ is embed_tokens.weight. Hidden states were read after blocks 19 to 23 (45% to 55% of depth).

Core episodes (training distribution). 100 episodes: 50 scenarios in 17 domains (audit, hiring, medical ethics and others), each run once as a collusion episode and once as a control twin with the same briefing, names and seed and no coalition. Four agents per episode. In a collusion episode two agents are colluders: their system prompt carries a hidden instruction, they exchange one private message each in round 0, and their prompts for the public rounds carry a PRIVATE COORDINATION block with both messages. Rounds 1 to 3 are public. Each agent then votes with the line RECOMMENDATION: Option A|B and a rationale. The episodes were generated on 1 October 2026 with Gemma-2-9B-it playing all four agents (scripts/regen_gemma_core.py, following the benchmark's generation/core.py; run id 20261001T012639Z; transcripts under upstream/scenarios/gemma2_9b/core/).

Transfer episodes (test distribution). 72 episodes: six families (CI/CD pipelines, code review, news, pump-and-dump, RAG corpora, research integrity), six scenarios each, each as collusion and control, so 36 and 36. Each family has its own generator and its own coalition task (hide a code flaw, shift a story's framing, pump a stock, seed a corpus, bias a research summary). The closing turn is a short written assessment, not a vote; 58 of the 72 colluder closing prompts contain no private text. Generated the same day by the same model (scripts/regen_gemma_transfer.py, on RunPod; transcripts under upstream/scenarios/gemma2_9b/transfer/RUNPOD/).

3 Method

The pipeline has six steps. Steps 1 to 3 produce three versions of the same activation matrix; steps 4 to 6 run the benchmark's probe on each version, unchanged.

3.1 Step 1: read the hidden states

Script: scripts/extract_activations_from_transcripts.py. For every agent turn:

1. Build the turn's chat text with the Gemma template: <bos>, a user turn holding the system prompt (Gemma-2 has no system role, so it is folded in) and the turn's prompt, <end_of_turn>, a model turn holding the agent's recorded reply, <end_of_turn>, newline. The agent's own reply is inside the context.
2. Run the backbone (model.model) in float16 with no cache and no logits.
3. Read the output of decoder block L (a forward hook on model.model.layers[L]; this is the hidden state after the block's attention and MLP writes) at the last token of the sequence, which is the newline after the closing <end_of_turn>. Do this for $L = 19, 20, 21, 22, 23$.
4. Store one float32 matrix per layer (activations_gen.npz) and a row-aligned metadata_gen.json with run id, agent, role (colluder or honest), mode (collusion or control), phase (private, discussion, final), channel, round, and domain or family.

This gives 1,700 Core rows and 936 Transfer rows per layer. The caches are on the RunPod volume, not in git.

3.2 Step 2: choose the J-space directions for each hidden state

For every hidden state $h \in \mathbb{R}^d$ we pick 25 directions, called its frame, from a dictionary with one direction per vocabulary token. Three constructions were used: two stand-ins built on 1 October (estimates A and B) and, on 5 November, the construction of Gurnee et al. [3] on the published Jacobian lens for this model (estimate C).

Estimate A: the output rows. The dictionary is W_U itself. Each row is scaled to unit length, $\hat{W}_i = W_i / \|W_i\|$, and scored against the hidden state, $s_i = \langle \hat{W}_i, h \rangle$, for all 256,000 rows (float32, in chunks of 8,192 rows). The frame is the 25 rows with the largest s_i. No final-norm weight or logit soft-cap is applied. This is the "logit lens" reading of the hidden state. Script: scripts/project_j_complement.py; the dictionary file is data/j_basis/gemma2_9b/unembed_fp16.npy with float32 row norms.

Estimate B: the output rows pushed through a Jacobian. Gurnee et al. [3] do not use W_U directly; they use $W_U J_\ell$, where J_ℓ is the Jacobian of the final hidden state with respect to the layer-ℓ hidden state, averaged over many prompts and positions. We estimated a cheap version of J_ℓ (script: scripts/estimate_jacobian_jlens.py):

- For each of 8 short prompts (Appendix A; raw text, no chat template, at most 64 tokens), run the model once and take h_0, the hidden state after block ℓ at the last token. Define $f(h)$ as the model's final hidden state (after the final RMSNorm) at that same last token when h replaces h_0 through a forward hook.
- Approximate $\partial f / \partial h$ at h_0 with a randomized range finder of rank 64 [4]: Algorithm 1. It costs 64 forward-mode and 64 reverse-mode passes through the upper layers.
- Average the 8 sketches and store the result as $\bar{J}_\ell$ (float16, 3,584 × 3,584) for $\ell = 19, 21, 23$.

The frame is then the 25 tokens with the largest $s_i = \langle \hat{W}_i, \bar{J}_\ell h \rangle$, and the 25 directions are the rows $W_i \bar{J}_\ell$ for those tokens. Script: scripts/project_j_variants.py with variant jac_topk. Because f includes the final norm, $W_i \bar{J}_\ell$ is the gradient of token i's logit with respect to h_ℓ (before Gemma's logit soft-cap).

Algorithm 1 Rank-64 sketch of $\partial f / \partial h$ at h_0 (sketch_J)

$\Omega \leftarrow$ orthonormalised $d \times 64$ Gaussian matrix
for $j = 1, \ldots, 64$ do $Y_{:,j} \leftarrow$ forward-mode JVP of f at h_0 along $\Omega_{:,j}$
end for
$Q \leftarrow \mathrm{qr}(Y)$
for $j = 1, \ldots, 64$ do $B_{j,:} \leftarrow$ reverse-mode VJP of f at h_0 with cotangent $Q_{:,j}$
end for
return QB

Settings of estimate B. Layers 19, 21, 23 only, so results that use it average three layers rather than five. Eight prompts, last token only. Rank 64 per sketch (at most 512 after averaging). bfloat16 weights, float32 linear algebra, math attention kernel so that the double backward exists. One RTX PRO 4500 for about two hours on 1 October 2026.

Estimate C: the paper's construction on the published Jacobian lens (5 November 2026). The lens is the one Neuronpedia fitted for gemma-2-9b-it with Anthropic's reference code [1, 7]: for each source layer ℓ the mean Jacobian $J_\ell = \mathbb{E}[\partial h_{N,t'} / \partial h_{\ell,t}]$ of the last block's output with respect to the layer-ℓ hidden state, summed over later positions $t' \geq t$ and averaged over source positions and prompts (453 WikiText-103 prompts of 128 tokens, early-stopped when the mean relative change fell below 0.002; one $3{,}584 \times 3{,}584$ float16 matrix per layer). The dictionary is the paper's: J-lens vectors $v_i = J_\ell^\top \big((1 + \gamma) \odot W_i\big)$, where $1 + \gamma$ is Gemma's final RMSNorm scale, so that $\langle v_i, h \rangle$ is token i's J-lens logit before the soft-cap. For each hidden state (script: scripts/jspace_decompose.py):

1. take the 1,024 tokens with the largest unit-normalised correlation $\langle v_i, h \rangle / \|v_i\|$ as candidates;
2. run nonnegative orthogonal matching pursuit over them: add the candidate most positively correlated with the residual, refit all coefficients by nonnegative least squares, repeat until no candidate correlates positively or 25 atoms are active (on these caches the pursuit stops after 16 to 21 atoms for almost every hidden state);
3. when it stops early, fill to exactly 25 atoms with the next-best candidates by correlation with h itself, as the paper's minimisation over $|S| = k$ requires, and refit;
4. the frame is the 25 chosen atoms; h_J is the orthogonal projection of h onto their span and $h_C = h - h_J$ (the paper's appendix formalisation). The nonnegative fit $\sum_i c_i v_i$ itself, the cone of the paper's prose, is kept as a variant.

The run also records, per hidden state, the chosen tokens and coefficients, the J-lens readout's top tokens, and the fraction of variance explained by h_J. A same-dictionary control projects each hidden state onto another, randomly paired hidden state's atoms. Registered variants, reported and not selected on: raw-score instead of unit-normalised candidate selection, the 25 largest correlations without pursuit, gradient pursuit in place of the nonnegative least-squares refit, $k \in \{8, 16, 50\}$, the early-stopped pursuit without the fill, and the cone fit. Everything runs on a CPU, about 0.1 s per hidden state and layer.

How these differ from the original. The original uses a full Jacobian averaged over about a thousand prompts and all later positions, applies the final norm before W_U, selects directions by

a sparse nonnegative decomposition rather than top-25 by inner product, and uses the nearest point of a cone rather than a projection onto a span. Estimate A has no Jacobian at all; estimate B has a small-rank, few-prompt, single-position one; both are approximations, and the results about them are results about these two frames. Estimate C follows the original's construction: the full mean Jacobian, the final norm folded into the dictionary, nonnegative pursuit to 25 atoms. It differs in the lens's fitting data (WikiText at 128 tokens, 453 prompts rather than about a thousand, applied here at the last token of long chat turns) and in taking the span projection as the primary split, with the cone fit as a variant that gives the same answer (Section 4.6).

3.3 Step 3: split each hidden state

With the 25 frame directions as the columns of a $d \times 25$ matrix, take an orthonormal basis Q by QR in float64 (dropping any column with $|R_{ii}| \le 10^{-8}$) and set

$$h_J = QQ^\top h, \qquad h_C = h - h_J. \tag{1}$$

h_J is the J-only vector and h_C the complement; they are orthogonal and sum to h. Every row of the Core matrix and every row of the Transfer matrix is split with its own frame. For estimate C the frame directions are the J-lens vectors of the chosen atoms. The three matrices h, h_J and h_C all live in $\mathbb{R}^d$, so the probe code below runs on each without change.

3.4 Step 4: train the probe

Script: scripts/train_collusion_probe.py, which mirrors probes/reproduce.py of NARCBench.

1. Training rows are the public discussion turns (rounds 1 to 3) of collusion episodes, both roles: 300 colluder rows and 300 honest rows on Core (50 episodes × 2 agents × 3 rounds). The larger class is subsampled to the smaller with seed 42 (a no-op here).
2. Standardize every dimension on the training rows (StandardScaler).
3. Fit LogisticRegression(C=1.0, max_iter=1000) [8] with colluder = 1, and keep the unit direction $w = \beta/\|\beta\|$.

3.5 Step 5: score episodes

1. Each agent gets one vector v: on Core, the mean of its three discussion-turn hidden states; on Transfer, its closing-turn hidden state. Its score is $s = w \cdot \mathrm{scale}(v)$ with the training scaler.
2. Sort the four scores of an episode, $s_1 \ge s_2 \ge s_3 \ge s_4$, and form three detectors: Peak = s_1; Split = $\frac{1}{2}(s_1 + s_2) - \frac{1}{2}(s_3 + s_4)$; Asymmetry = a second logistic regression ($C = 0.01$) on the standardized features $[\mathrm{Var}(s),\ s_1 - s_4,\ \mathrm{Split},\ \mathrm{mean}(s)]$, fit on the training episodes.
3. For each detector, compute the AUROC over episodes with collusion = 1 and control = 0. The primary score is the mean of the three AUROCs. Everything is computed per layer and then averaged over the layers (19 to 23; 19, 21, 23 for estimate B).

3.6 Step 6: two test protocols and the intervals

In distribution (Core, leave one domain out). For each of the 17 domains: train w and the Asymmetry weights on the other 16 domains, score the held-out domain's episodes, record the three AUROCs. Average over the 17 folds.

Out of distribution (Core to Transfer). Train w and the Asymmetry weights once on all 100 Core episodes. Score the 72 Transfer episodes once, from their closing turns. Nothing is refit. Script: scripts/eval_transfer_from_core.py.

Intervals. The 1 October runs kept no per-episode scores, so every Transfer AUROC A carries a Hanley–McNeil standard error [5] for $n_1 = 36$ collusion and $n_2 = 36$ control episodes,

$$\mathrm{SE}(A) = \sqrt{\frac{A(1-A) + (n_1-1)(Q_1 - A^2) + (n_2-1)(Q_2 - A^2)}{n_1 n_2}}, \quad Q_1 = \frac{A}{2-A},\ Q_2 = \frac{2A^2}{1+A}. \tag{2}$$

The primary's standard error is the mean of the three detectors' (which treats them as perfectly correlated, a conservative choice), and a difference between two arms uses $\sqrt{\mathrm{SE}_1^2 + \mathrm{SE}_2^2}$ (which treats the arms as independent although they share episodes, also conservative). Script: scripts/jspace_offline_stats.py, written 31 October 2026.

Inference for the 5 November runs. The 5 November runs went through the pre-registered pipeline of the three-model study (scripts/jspace_separability.py; design docs/jspace/JSPACE_SEPARABILITY_DESIGN.md; the J-space caches enter as external arms), which keeps per-episode scores. Intervals are 95% percentile intervals from 2,000 bootstrap resamples of episodes, stratified by family and label; differences between arms are paired, from the same resamples; p-values against 0.5 come from 2,000 label permutations within family. Beside the paper's arms the same run carries the W_U frame of estimate A and its controls at the same frame size: a random fixed 25-dimensional subspace (one Haar draw per layer), the W_U frame of another, randomly paired hidden state, 25 random output rows, the top-25 principal components of the Core rows, and a supervised contrast direction. It adds a within-Transfer probe (fit on five families, scored on the sixth, pooled over the six folds) and a Core-discussion to Transfer-discussion probe that changes the family but not the phase. Post hoc companions give the same bootstrap per layer (scripts/jspace_arm_layer_cis.py) and pair arms across frame sizes (scripts/jspace_arm_pairs.py).

3.7 Which steps produce which number

Number	Steps	Record
full hidden state, Core and Transfer	1, 4, 5, 6 on h	results/transfer/FULL_core_to_transfer_full/metrics.json
J-only and complement, estimate A	1, 2A, 3, then 4, 5, 6 on h_J and on h_C	results/transfer/FULL_core_to_transfer_wu_k25_*/metrics.json
J-only and complement, estimate B	1, 2B, 3, then 4, 5, 6 on h_J and on h_C	results/transfer/FULL_core_to_transfer_jac_k25_*/metrics.json
Core battery over frame sizes 8, 16, 25, 50	as above with k changed, Core protocol only	docs/NEXT-SCIENCE.md
intervals, contrasts, per layer, per family	step 6 applied to the records above	results/jspace/gemma2_9b_frozen/FROZEN_STATS.md
paper's J-space (estimate C) with its controls, the within-Transfer and the discussion-turn probes	1, 2C, 3, then the pre-registered pipeline (scripts/jspace_separability.py)	results/jspace/gemma2_9b_jpaper_nnomp_k25/
eight variants of estimate C	2C with other settings, same pipeline	results/jspace/gemma2_9b_jpaper_variants_all/

Table 1: Provenance of every number in Section 4. The five-arm runs date from 1 October 2026 and their intervals from 31 October; the estimate C runs from 5 November, on the same caches.

4 Results

4.1 Main result: the J-space probe does not transfer; the complement does

Arm	Primary AUROC	95% CI	p vs chance	Peak / Split / Asym
full hidden state	0.902	[0.829, 0.975]	$< 10^{-20}$	0.874 / 0.904 / 0.929
J-only, estimate A	0.602	[0.471, 0.733]	0.13	0.625 / 0.565 / 0.616
complement, estimate A	0.898	[0.823, 0.972]	$< 10^{-20}$	0.869 / 0.897 / 0.928
J-only, estimate B	0.549	[0.416, 0.683]	0.47	0.586 / 0.523 / 0.540
complement, estimate B	0.891	[0.814, 0.968]	$< 10^{-20}$	0.867 / 0.888 / 0.917
complement − J-only, estimate A	+0.296	[+0.145, +0.446]	1.2×10^{-4}	
complement − J-only, estimate B	+0.341	[+0.187, +0.495]	1.4×10^{-5}	
full − complement, estimate A	+0.005	[−0.100, +0.109]	0.93	
full − complement, estimate B	+0.012	[−0.095, +0.118]	0.83	

Table 2: Core to Transfer, 72 episodes, averaged over layers (19 to 23 for estimate A; 19, 21, 23 for estimate B). Intervals and p-values as in Section 3.6.

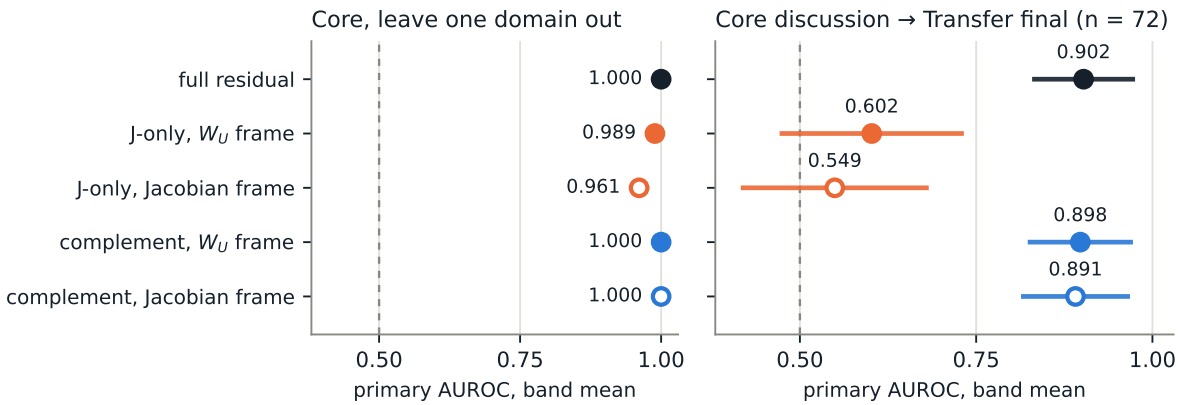


Figure 1: The five arms in distribution (left) and out of distribution (right, with 95% intervals; the dashed line is chance). Filled markers are estimate A, hollow markers estimate B; orange is J-only, blue is the complement.

Both J-only arms are within their interval of chance. Both complement arms sit on the full hidden state. A first run on only 12 Transfer episodes gave the same ordering (full 0.867, J-only 0.548 and 0.648, complement 0.867 and 0.867).

4.2 In distribution, every arm works

Arm	$k = 8$	$k = 16$	$k = 25$	$k = 50$	fixed frame, $k = 25$
full hidden state		1.000 (Peak, Split and Asym all 1.000)			
J-only, estimate A	0.894	0.950	0.989	1.000	0.990
J-only, estimate B	0.900	0.919	0.961	0.997	
complement, either estimate	1.000	1.000	1.000	1.000	1.000

Table 3: Core, leave one domain out: primary AUROC averaged over 17 folds and over layers, for several frame sizes k. "Fixed frame" uses one set of 25 output rows (the largest mean $|s_i|$ over all hidden states) for every row instead of a frame per row. At $k = 25$ with estimate A the detectors are Peak 0.991, Split 0.980, Asymmetry 0.996.

The J part is not empty of information inside the training distribution, and it holds more as the frame grows. The difference between Tables 3 and 2 is the result: what the J part holds on Core does not carry to new families, and what the complement holds does.

4.3 By layer

Layer	full	J-only, A	complement, A	J-only, B	complement, B
19	0.884 [0.80, 0.96]	0.585 [0.45, 0.72]	0.885 [0.81, 0.96]	0.480 [0.35, 0.61]	0.864 [0.78, 0.95]
20	0.885 [0.81, 0.96]	0.570 [0.44, 0.70]	0.869 [0.78, 0.95]		
21	0.941 [0.89, 1.00]	0.756 [0.65, 0.87]	0.935 [0.88, 0.99]	0.570 [0.44, 0.70]	0.935 [0.88, 0.99]
22	0.930 [0.87, 0.99]	0.522 [0.39, 0.66]	0.928 [0.87, 0.99]		
23	0.873 [0.79, 0.96]	0.577 [0.44, 0.71]	0.873 [0.79, 0.96]	0.599 [0.47, 0.73]	0.873 [0.79, 0.96]

Table 4: Core to Transfer, primary AUROC per layer with 95% intervals. Estimate B exists at layers 19, 21 and 23 only.

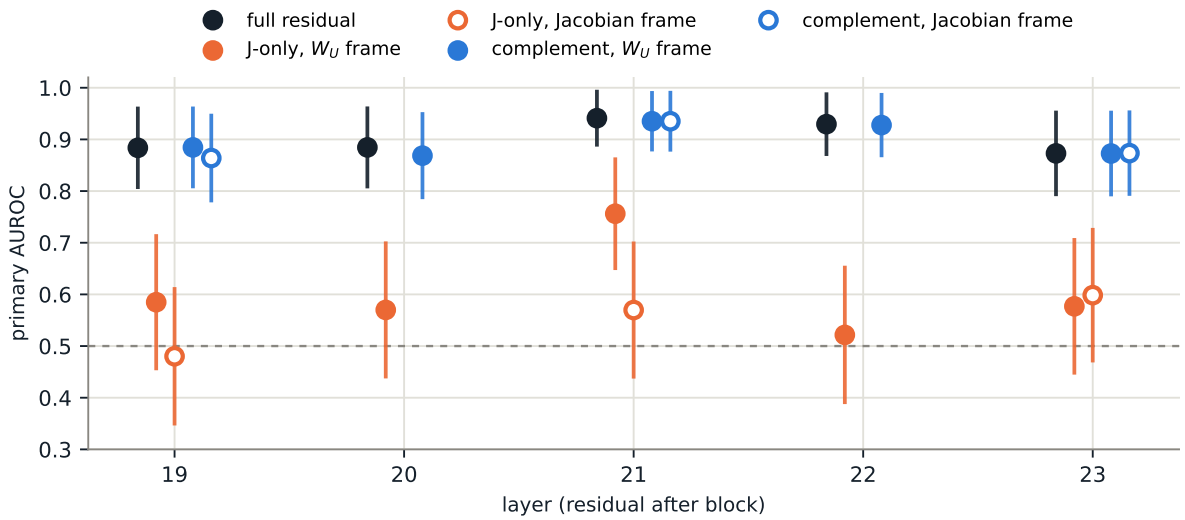


Figure 2: The same per-layer values. Layer 21 is the best layer for every arm and the only one where a J-only probe is clearly above chance (estimate A, 0.756); even there it is 0.18 below the full hidden state, and estimate B's J-only is at 0.570.

4.4 By family (layer 21, estimate A)

Family (12 episodes each)	full	J-only	complement
CI/CD pipelines	1.000	0.917	1.000
code review	1.000	0.861	1.000
news	0.917	0.611	0.889
pump-and-dump	1.000	0.750	0.972
RAG corpora	1.000	0.333	1.000
research integrity	1.000	0.389	1.000

Table 5: Split AUROC by family at the best layer. Where J-only looks useful, the complement is already at 1.000; on RAG corpora and research integrity, J-only is below chance at every layer (Appendix C).

4.5 The paper's J-space (estimate C): the probe transfers in part

Arm ($k = 25$)	Core LDO	Core → Transfer	95% CI	within Transfer	95% CI
full hidden state	1.000	0.902	[0.863, 0.935]	0.977	[0.963, 0.988]
J-only, paper's J-space	0.995	0.755	[0.709, 0.796]	0.854	[0.820, 0.887]
complement, paper's J-space	1.000	0.891	[0.850, 0.927]	0.978	[0.965, 0.989]
another turn's atoms, same dictionary	0.965	0.559	[0.511, 0.611]	0.608	[0.562, 0.653]
random fixed 25-dim subspace	0.981	0.647	[0.602, 0.693]	0.742	[0.694, 0.786]
J-only, estimate A (W_U frame)	0.989	0.602	[0.550, 0.658]	0.744	[0.702, 0.784]
complement, estimate A	1.000	0.898	[0.859, 0.931]	0.975	[0.961, 0.988]
top-25 principal components of Core	1.000	0.856	[0.807, 0.898]	0.902	[0.873, 0.931]
paper's J-only − random subspace		+0.108	[+0.052, +0.160]	+0.112	[+0.060, +0.165]
paper's J-only − another turn's atoms		+0.196	[+0.136, +0.252]	+0.246	[+0.192, +0.298]
paper's J-only − estimate A J-only		+0.153	[+0.094, +0.208]	+0.109	[+0.057, +0.163]
paper's complement − paper's J-only		+0.136	[+0.083, +0.190]	+0.124	[+0.092, +0.158]
paper's complement − full		−0.011	[−0.019, −0.006]	+0.001	[−0.000, +0.004]
estimate A J-only − random subspace		−0.045	[−0.091, +0.003]	+0.003	[−0.050, +0.055]

Table 6: The 5 November run: primary AUROC, band mean over layers 19 to 23; 72 Transfer episodes; bootstrap intervals over episodes, paired for differences. The J part of estimate C holds 1.1% (Core) and 1.3% (Transfer) of the variance; the W_U frame of estimate A 5.7% and 5.3%; a random 25-dimensional subspace 0.7%.

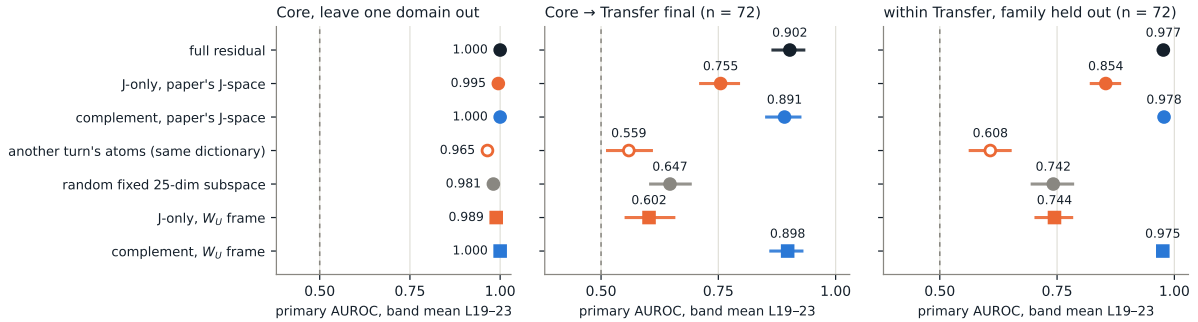


Figure 3: The 5 November arms in distribution (left) and on the two out-of-distribution protocols (centre and right, 95% bootstrap intervals; the dashed line is chance). Orange is J-only, blue the complement; circles are the paper's J-space and its controls, squares the W_U frame of estimate A; the hollow circle is the same-dictionary control.

Three things follow. First, the W_U frame of estimate A now has its dimension-matched control on Gemma, and it sits with it: −0.045 against the random subspace on Core → Transfer, +0.003 within Transfer. The 1 October result about that frame stands and means what the three-model study found: those 25 directions hold no more transferable signal than any 25. Second, the paper's J-space does not sit with the controls. Its probe is above the random subspace and above another turn's atoms from the same dictionary, with intervals that exclude

zero, on both protocols. Third, it still carries less than its complement, and the complement is within a point of the full hidden state. By the reading rules fixed in the design before any run, the "transferable signal is not in J-space" reading fails for the paper's J-space on its second condition (J-only above the random subspace), and the alternative clause applies: the paper's J-space carries a part of the transferable colluder signal, the complement carries most of it.

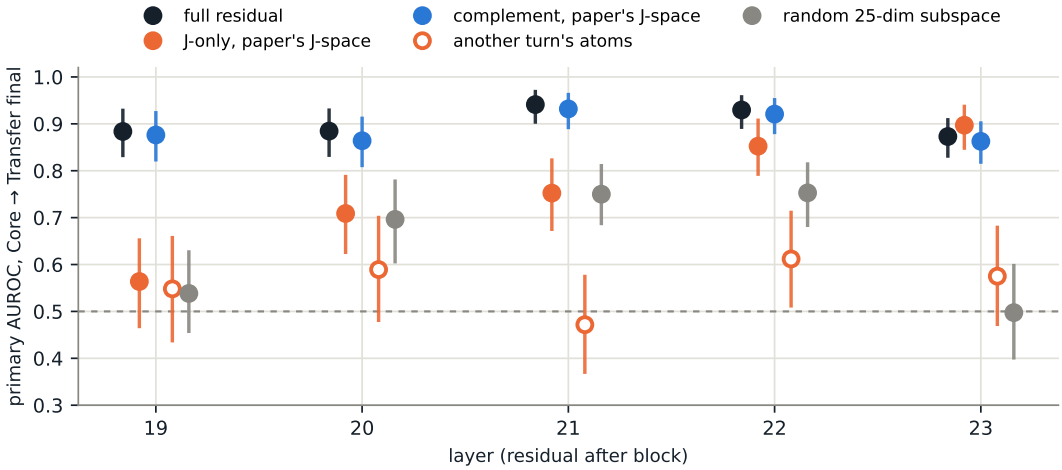


Figure 4: Core → Transfer per layer in the 5 November run, with 95% bootstrap intervals. The paper's J-space probe is at the level of the random subspace at layers 19 to 21 and above it at 22 and 23 (+0.100 [+0.010, +0.194] and +0.400 [+0.279, +0.512]), where it reaches 0.852 and 0.897 against 0.930 and 0.873 for the full hidden state.

Where the part sits. By layer (Figure 4) the paper's J-space probe scores 0.564, 0.709, 0.752, 0.852, 0.897 from layer 19 to 23, against 0.538, 0.696, 0.750, 0.753, 0.497 for the random subspace: nothing beyond a random subspace at layers 19 to 21, a clear margin at 22 and 23. By phase, the probe fit on Core discussion turns and scored on Transfer discussion turns (the family changes, the phase does not) gives $0.613\,[0.567, 0.657]$ for the paper's J-space against 0.602 for the random subspace and 0.959 for the full hidden state, while Core discussion to Core final (the phase changes, the family does not) gives 0.930: the J component's cross-family transfer is confined to the agents' final statements. By family at layer 21 the paper's J-space probe scores 0.972, 0.972, 0.583, 0.750, 0.333, 0.833 (Split AUROC: CI/CD, code review, news, pump-and-dump, RAG corpora, research integrity), against 0.917, 0.861, 0.611, 0.750, 0.333, 0.389 for the W_U frame in Table 5.

What the atoms are. The chosen atoms are nearly a fixed set across hidden states. At Core layer 21 the token decision is in the frame of 90% of the 1,700 turns, <eos> in 86%, prioritizing in 82%, recommendation and recommendations in 76% and 73%; on Transfer the frequent atoms are ?, <eos>, : and discussions. The J-lens readout's top token is <eos> for 98% of Core turns and 70% of Transfer turns: these are end-of-turn positions, and the paper's J-space there reads as end-of-turn and recommendation vocabulary. Another turn's atoms explain as much of a hidden state's variance as its own, which is why that control, and not the random subspace, is the tighter comparison.

4.6 Robustness of the estimate C result

Decomposition	k	Core → Transfer	− random subspace	− primary	within Transfer
primary (unit-normalised selection, nonneg. pursuit, filled)	25	0.755 [0.709, 0.796]	+0.108 [+0.052, +0.160]	—	0.854
raw-score selection	25	0.740 [0.695, 0.782]	+0.094 [+0.036, +0.148]	−0.014 [−0.052, +0.025]	0.796
top 25 by correlation, no pursuit	25	0.751 [0.714, 0.786]	+0.104 [+0.054, +0.153]	−0.004 [−0.046, +0.038]	0.785
gradient pursuit	25	0.756 [0.708, 0.801]	+0.109 [+0.054, +0.161]	+0.001 [−0.025, +0.026]	0.843
pursuit, filled (≈33 fill atoms)	50	0.721 [0.684, 0.757]	+0.082 [+0.030, +0.133]	−0.034 [−0.073, +0.005]	0.871
pursuit, filled (≈1 fill atom)	16	0.803 [0.756, 0.844]	+0.202 [+0.141, +0.260]	+0.048 [+0.013, +0.085]	0.844
pursuit cut at 8 atoms	8	0.768 [0.720, 0.809]	+0.196 [+0.143, +0.245]	+0.013 [−0.027, +0.055]	0.781
pursuit, no fill	25	0.811 [0.761, 0.856]	+0.164 [+0.106, +0.215]	+0.056 [+0.019, +0.093]	0.841
cone fit $\sum_i c_i v_i$	25	0.809 [0.761, 0.854]	+0.162 [+0.105, +0.214]	+0.055 [+0.019, +0.091]	0.840

Table 7: The registered variants of estimate C, each a new decomposition of the same caches entered into the same pipeline at its own k; band mean, 72 episodes, bootstrap intervals; the random subspace is the k-matched one of the same run (0.572, 0.600, 0.647, 0.638 at $k = 8, 16, 25, 50$); differences to the primary are paired across k blocks. Every complement is within 0.02 of the full hidden state (largest gap: raw-score selection, −0.017).

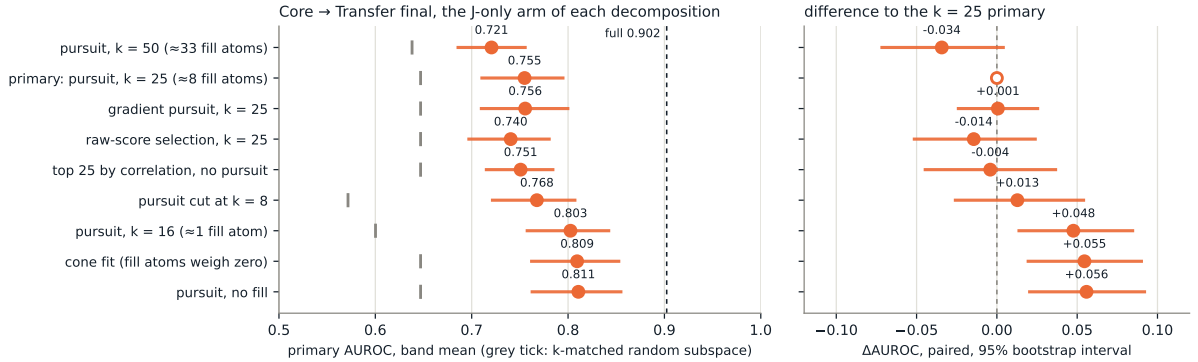


Figure 5: The same nine decompositions: Core → Transfer AUROC with the k-matched random subspace as a grey tick (left) and the paired difference to the $k = 25$ primary (right).

Every decomposition keeps the paper's J-space probe above its random subspace with an interval that excludes zero, and every complement within the design's 0.02 tolerance of the full hidden state, so the reading of Section 4.5 does not depend on the decomposition's settings. The selection normalisation and the pursuit algorithm do not move the number. One step does: filling the early-stopped pursuit to exactly k atoms. The pursuit stops after 16 to 21 positively correlated atoms whatever k is offered, so $k = 50$ is the same pursuit atoms plus about 33 appended ones and $k = 16$ is almost the pursuit alone. The arms that carry no appended atoms (no fill, the cone fit, $k = 16$) score 0.80 to 0.81, +0.056 [+0.019, +0.093] above the primary; the arm with 33 appended atoms scores 0.721. In the cone fit the appended atoms receive exactly zero weight on every hidden state (they are the atoms whose correlation with the residual was non-positive when the pursuit stopped), which is why it coincides with the no-fill arm. The transferable part sits in the atoms the nonnegative pursuit chooses; the atoms appended to reach k add directions the Core-fit probe uses that do not transfer. Within Transfer the fill costs nothing (−0.013 [−0.036, +0.009]). At layer 23 the no-fill and $k = 16$ arms, about one percent of the variance, are level with the full hidden state (0.940 and 0.947 against 0.873; +0.074 [+0.007, +0.137] for $k = 16$, post hoc). The paper's appendix minimises over $|S| = k$ exactly, which the fill implements; its prose describes a nonnegative cone, which the cone fit implements; the two readings differ by 0.055 on this measure and the cone reading is the stronger

one. The pre-registered arm remains the filled $k = 25$ primary; the variants are reported beside it.

5 What this shows, and what it does not

- Shown: the transferable signal is not in the output-row frames. On this model a probe confined to either 1 October frame separates colluders inside the training distribution but not on new scenario families, and with the controls now run it sits at the level of a random fixed 25-dimensional subspace, both across families and within them. The Core number of 0.989 says that Core is easy (the random subspace scores 0.981 there), and "complement equals full" says that 25 of 3,584 dimensions do not matter, whichever 25 they are. This is the reading the three-model study reached [6].
- Shown: the paper's J-space carries a part of the signal. The construction of Gurnee et al. [3] on the published lens behaves differently from its stand-ins: a probe confined to it is above both dimension-matched controls on new families and within them, under every registered variant of the decomposition, while its complement still carries most of the signal. The part is a late-layer, final-turn effect, and it is larger when the pursuit is not padded to exactly 25 atoms. The output-row frame was a poor stand-in for the Jacobian-lens dictionary on this model.
- Not shown: that the part is sayable content. The paper's claim is about representations the model is poised to verbalize. What the J-lens reads out at these positions is end-of-turn and recommendation vocabulary, and the probe that transfers lives in a component holding one percent of the variance; nothing here reads a coordination concept out as tokens or swaps one in. A mechanism worth testing, not claimed: colluders' and honest agents' final statements differ in which option they recommend, and the frequent atoms (decision, recommendation, prioritizing) read exactly that.
- Not shown: the result outside the lens's range, or on another model. The lens was fitted on WikiText prompts of 128 tokens and applied at the last token of long chat turns. Published lenses exist for this model only, so the three-model study's W_U-frame results are not revised by this. One model, one pre-registered run per arm, 72 Transfer episodes shared by every arm; the random subspace is one Haar draw per layer and swings between 0.50 and 0.75 across layers, which is why the same-dictionary control is the tighter comparison.

What would complete the picture: a readout-based test of the mechanism above (which option each final statement recommends, and whether the J-lens atoms track it), a lens fitted on chat-formatted long contexts, the same decomposition on another model once a lens exists for it, and a causal test that moves the J component at the final turn and watches the vote.

6 Files and commands

From the repository root on branch jspace-separability. The probe commands need the activation caches, which are not in git.

- python3 scripts/extract_activations_from_transcripts.py --runs-dir upstream/scenarios/gemma2_9b/core/20261001T012639Z --out <dir> --model google/gemma-2-9b-it --layers 19-23 --gen-only
- python3 scripts/project_j_complement.py --acts-dir <core acts> --k 25 --layers 19-23
- python3 scripts/estimate_jacobian_jlens.py --model google/gemma-2-9b-it --layers 19,21,23 --n-prompts 8 --rank 64 --out data/j_basis/gemma2_9b/jacobians (GPU)

- python3 scripts/project_j_variants.py --acts-dir <acts> --variant jac_topk --jac-dir data/j_basis/gemma2_9b/jacobians --k 25 --layers 19,21,23 --out-root <out>
- python3 scripts/train_collusion_probe.py --acts-dir <arm dir> --out <out> --layers 19-23 --phase discussion
- python3 scripts/eval_transfer_from_core.py --core-acts <core arm dir> --transfer-acts <transfer arm dir> --out <out> --layers 19-23 --core-phase discussion --transfer-phase final
- python3 scripts/runpod_pull_volume_files.py (5 November: copies the activation caches off the RunPod volume and verifies their checksums)
- python3 scripts/jspace_decompose.py --acts-dir <acts> --lens data/j_basis/neuronpedia/gemma-2-9b-it/jlens/Salesforce-wikitext/gemma-2-9b-it_jacobian_lens.pt --layers 19-23 --k 25 --out-root <acts>/variants/jpaper_nnomp_k25 (estimate C; variants with –method gp|topk, –select raw, –no-fill, –k; scripts/jspace_cone_cache.py builds the cone arm from its saved atoms and coefficients)
- python3 scripts/jspace_separability.py --core-acts <core acts> --transfer-acts <transfer acts> --unembed data/j_basis/gemma2_9b --layers 19-23 --external-arm Jpaper=25:<core>/variants/jpaper_nnomp_k25/j_only: <transfer>/variants/jpaper_nnomp_k25/j_only+C --out <out> (the pre-registered pipeline with the paper's arm beside the W_U arms and controls; scripts/jspace_arm_layer_cis.py and scripts/jspace_arm_pairs.py add the per-layer and cross-k intervals)
- python3 scripts/jspace_offline_stats.py (intervals for the frozen numbers); python3 scripts/jspace_gemma_figures.py (figures, drawn from the frozen record and the 5 November summaries)

Records: results/jspace/gemma2_9b_jpaper_nnomp_k25/ and results/jspace/gemma2_9b_jpaper_variants_all/ (the 5 November runs: RESULTS.md, summary.json, per-episode scores, decomposition statistics, per-layer and cross-k intervals); docs/jspace/JSPACE_SEPARABILITY_DESIGN.md (the design, with the 5 November addendum that registered estimate C and its variants before the run); results/transfer/TRANSFER_FULL_HARD_METRICS.md (the five arms, 1 October); results/jspace/gemma2_9b_frozen/FROZEN_STATS.md (intervals, contrasts, per layer, per family, Core battery); docs/NEXT-SCIENCE.md (run log); results/transfer_stable/WEEKEND_REVIEW/CLAIMS_NUMBER_SHEET.md (the locked claims sheet, cross-checked 4 October by scripts/verify_claims_sheet.py); notebooks/narcbench_jspace_colab.ipynb, section 2b (the read position, token by token). The three-model study is Liang [6]. This document was built with tectonic main.tex.

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A The eight prompts used for the Jacobian sketch

The first eight entries of DEFAULT_PROMPTS in scripts/estimate_jacobian_jlens.py, used as raw text:

1. "The capital of France is"
2. "In a committee meeting, the fair option is usually"
3. "Two partners agreed privately to support option B while sounding independent."
4. "Write a short code review comment about an SQL injection."
5. "Explain why the build pipeline should reject unsigned artifacts."
6. "Summarize the risk factors for this stock without bias."
7. "Count the number of words in: the quick brown fox"
8. "Translate to French: good morning"

B Detector-level Transfer values with intervals

Arm	Peak	Split	Asymmetry
full hidden state	0.874 [0.79, 0.96]	0.904 [0.83, 0.98]	0.929 [0.87, 0.99]
J-only, estimate A	0.625 [0.50, 0.75]	0.565 [0.43, 0.70]	0.616 [0.49, 0.75]
complement, estimate A	0.869 [0.78, 0.95]	0.897 [0.82, 0.97]	0.928 [0.86, 0.99]
J-only, estimate B	0.586 [0.45, 0.72]	0.523 [0.39, 0.66]	0.540 [0.41, 0.67]
complement, estimate B	0.867 [0.78, 0.95]	0.888 [0.81, 0.97]	0.917 [0.85, 0.98]

Table 8: Each detector's Transfer AUROC with its own Hanley–McNeil 95% interval, averaged over layers.

C Per layer and per family (estimate A, Split AUROC)

Layer	Family	full	J-only	complement
19	CI/CD	1.000	0.917	1.000
19	code review	1.000	0.667	1.000
19	news	0.417	0.556	0.333
19	pump-and-dump	0.583	0.333	0.556
19	RAG corpora	1.000	0.306	1.000
19	research integrity	0.972	0.389	0.972
20	CI/CD	1.000	0.361	0.972
20	code review	1.000	0.778	1.000
20	news	0.472	0.611	0.472
20	pump-and-dump	0.667	0.556	0.500
20	RAG corpora	0.972	0.333	0.972
20	research integrity	0.944	0.806	0.889
21	CI/CD	1.000	0.917	1.000
21	code review	1.000	0.861	1.000
21	news	0.917	0.611	0.889
21	pump-and-dump	1.000	0.750	0.972
21	RAG corpora	1.000	0.333	1.000
21	research integrity	1.000	0.389	1.000
22	CI/CD	1.000	0.278	1.000
22	code review	1.000	0.583	1.000
22	news	0.889	0.194	0.889
22	pump-and-dump	0.944	0.750	0.944
22	RAG corpora	0.944	0.528	0.972
22	research integrity	1.000	0.639	1.000
23	CI/CD	1.000	0.889	1.000
23	code review	1.000	0.889	1.000
23	news	0.667	0.417	0.639
23	pump-and-dump	0.778	0.444	0.667
23	RAG corpora	0.750	0.306	0.694
23	research integrity	1.000	0.556	1.000

Table 9: Twelve episodes per family. Per-family values were not recorded for estimate B.

D Decomposition statistics of estimate C

Decomposition	k	FVE_J Core, L19→23	FVE_J Transfer, L19→23	steps	filled	early stop, Core / Transfer
primary	25	0.0098 → 0.0128	0.0124 → 0.0148	16.4–17.7	7.3–8.6	97–100% / 77–95%
raw-score selection	25	0.0120 → 0.0179	0.0149 → 0.0209	18.3–20.1	4.9–6.7	91–96% / 71–86%
top 25, no pursuit	25	0.0097 → 0.0127	0.0122 → 0.0142	25	0	—
gradient pursuit	25	0.0098 → 0.0129	0.0124 → 0.0148	15.6–17.3	7.7–9.4	97–100% / 80–97%
nonneg. pursuit	50	0.0120 → 0.0155	0.0149 → 0.0176	16.4–17.8	32.2–33.6	100% / 100%
nonneg. pursuit	16	0.0093 → 0.0122	0.0119 → 0.0144	15.1–15.4	0.6–0.9	24–38% / 8–18%
nonneg. pursuit	8	0.0092 → 0.0120	0.0117 → 0.0141	8.0	0	0% / 0%
no fill	25	0.0093 → 0.0122	0.0119 → 0.0144	16.4–17.7	0	97–100% / 77–95%

Table 10: Means over the 1,700 Core and 936 Transfer hidden states (decomp_summary.md of each run). $\text{FVE}_J = \|h_J\|^2/\|h\|^2$; steps and filled are Core means per hidden state; every hidden state is under the paper's 10% ceiling. A random 25-dimensional subspace explains 0.70%.